

Arvind P G

Senior Engineer

Rotterdam, Netherlands • +31 616749008 • p.arvindgovind@gmail.com

On Dependent Highly Skilled Migrant (HSM), no employer visa sponsorship required.

Python Engineer with 7+ years of experience in high-performance API design, scalable infrastructure, and cybersecurity tooling. Experienced with Django and FastAPI for building microservices and backend systems. Known for consistently debugging complex production issues with precision, delivering secure and optimized solutions, integrating diverse toolsets, and mentoring teams. Strong background in Docker, Git, Linux, PostgreSQL, and Kubernetes, with a focus on clean, tested code and technical rigor.

WORK EXPERIENCE

Senior Engineer

Dec 2022 – Present

FireCompass, India

- Managed backend development for an External Attack Surface Management platform using Python, Django, PostgreSQL, and Docker, delivering high-performance Django REST APIs by optimizing PostgreSQL queries with indexing and restructuring, achieving a 10x improvement in response times.
- Developed FastAPI-based microservices using SQLAlchemy ORM for database operations, and built a proxy service to orchestrate Playwright containers for automated browser-based security testing.
- Architected a global server-side Redis cache for API microservices, designing configurable caching policies and invalidation rules that balanced real-time data accuracy against latency requirements.
- Developed a versatile system that enables security engineers to add custom scripts, automatically converting their outputs into Knowledge Graphs and storing them in PostgreSQL alongside other security tool data, reducing data integration time from 2-3 days to hours, a 5x improvement.
- Enhanced asset monitoring workflows via randomized sampling, verification, and validation layers. Shifted metadata storage to Google Bigtable and scan data to Google Cloud Storage, improving reliability and consistency of asset updates.
- Engineered a robust system to manage and gracefully abort concurrent security scans by integrating with Google Pub/Sub for real-time abort signals, ensuring minimal impact on client networks, preventing unnecessary traffic.
- Integrated over 30 security tools, including Nmap and Nessus, to automate asset discovery, network and web vulnerability scanning achieving a 40% reduction in manual effort.
- Ensured the reliability of critical backend REST APIs by implementing comprehensive unit tests, significantly reducing production issues by 50%.
- Created and deployed a company-wide Slack alerting library that enables all micro-services to send real-time incident notifications, improving cross-team communication and accelerating incident response times.
- Enhanced a Day 1 CVE system using Airtable to streamline matching CVE data with affected assets in OpenSearch.
- Delivered security dashboards using ReactJs to display scan and CVE results in real-time, enhancing user decision-making.

Senior Full Stack Engineer

Jan 2020 – Dec 2022

VuNet Systems, India

- Re-architected access control for a Django-based Big Data Analytics Platform, integrating LDAP, ADFS SSO (Azure), and OAuth2, resulting in smoother onboarding and improved RBAC consistency across clients.
- Designed and implemented an HR Management System tied into platform authentication, reducing administrative friction and unifying role-based access control.
- Created a Node.js based monitoring tool using Google Puppeteer to perform passive application monitoring, automating performance tracking and issue detection for web applications across multiple clients.

- Reverse engineered Kibana to upgrade Elasticsearch version 6.8 to 7.17 to ensure compatibility.
- Collaborated with security audit teams to resolve vulnerabilities and strengthen platform security posture.
- Improved reporting capabilities by enabling PDF/CSV export through PhantomJS, reducing manual report generation effort.

Analyst

June 2018 – Dec 2019

Ernst & Young Associates LLP, India

- Enhanced an internal Django application by automating baseline configuration checks using Ansible, reducing manual audit effort.
- Conducted VAPT across web, Android, and iOS applications; reviewed secure configurations of network devices; supported DDoS testing of load balancers to harden client infrastructure.

EDUCATION

- **Bachelor of Engineering:** Computer Science And Engineering Aug-2014 - May-2018
 - Anna University - Chennai, India CGPA - 8.4
- **Senior Secondary School:** Computer Science May-2012 - Apr-2014
 - SBOA MHSS - Chennai, India 97.5%
- **Higher Secondary School:** May-2010 - Apr-2012
 - SBOA MHSS - Chennai, India 97.2% (Top 3)

SKILLS & OTHER

- Backend: Python (type-annotated), Django, FastAPI, Pydantic, Mypy
- Cloud & Infra: GCP, AWS, Docker, Kubernetes, GitHub CI/CD
- Databases: PostgreSQL, MySQL, MongoDB, Elasticsearch, Google Bigtable, Redis
- Frontend: React.js, TypeScript, AngularJS
- Security Tools: Burp Suite, Nmap, Nessus, Metasploit, Wireshark
- Other: Node.js, Puppeteer, RabbitMQ, Pytest

CERTIFICATIONS

- **Data Structures & Algorithms** June 2024
Cleared the skill mastery assessment by Scaler & InterviewBit (NSDC-certified).
- **Databases and SQL** August 2024
Cleared the skill mastery assessment by Scaler & InterviewBit (NSDC-certified).